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31 January 1984
SUPERSEDING
MIL-P-6131B
15 July 1969

MILITARY SPECIFICATION

PLUGS, DEHYDRATOR, INDICATING HUMIDITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for humidity indicating, dehydrator plugs that are intended to protect aircraft engine cylinders and other similar equipment against corrosion due to moisture and to indicate through color change unsafe levels of humidity (see 6.1).

1.2 Classification. Dehydrator plugs shall be of the following types, as specified (see 6.2).

Type I - Reusable.

Type II - Reusable and refillable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

Federal (Continued)

PPP-B-636	Box, Shipping, Fiberboard
PPP-C-96	Cans, Metal, 28 Gage and Lighter

Military

MIL-D-3716	Desiccants, Activated: For Dynamic Dehumidification
MIL-D-6054	Drum, Metal; Shipping and Storage
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes

STANDARDS

Federal

FED. TEST METHOD STD. NO. 101	Test Procedures for Packaging Materials

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MS3396	Plug, Dehydrator, Humidity Indicating, Engine Cylinder (14mm Thread)
MS3398	Card-Humidity-Indicating Color Comparison
MS27215	Plug, Dehydrator, Humidity Indicating, Engine Cylinder (18mm Thread)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of conflict between the text in this specification and references cited herein, the text of the specification shall take precedence.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society For Testing And Materials (ASTM)

ASTM D 1748 - Rust protection by metal preservation in the humidity cabinet.

ASTM D 3951 - Standard practice for commerical packaging. .

Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Plugs. The dehyrdator plugs shall be manufactured from any metal or plastic material or combination with a transparent plastic or glass viewing area. Materials used shall be of high quality and shall not be deleteriously affected by gasoline or other petroleum derivatives. (See 4.4.1 and table III.)

3.1.2 Gasket, exterior seating. Gasket, exterior seating, material for use on type I or II dehydrator plugs shall conform to MIL-R-6855, class 4, grade 80. (See 4.4.1 and table III.)

3.1.3 Sealing plug. The seal material shall protect the desiccant in the plug from moisture and be resuable without the aid of separately applied sealing compounds. (See 4.4.1 and table III.)

3.1.4 Retainer (filter). The retainer (filter) shall be made of any material (nominally metal) that can be formed with such openings to retain the desiccant granuales and to allow the passage of watervapor at the rate given in table III for desiccant watervapor absorption. (See 4.4.1.)

3.2 Design.

3.2.1 Plugs. The plugs shall conform to MS3396 or MS27215.

3.2.1.1 Type I. Type I plug shall be considered reusable provided it meets all requirements of the specification except those that apply specifically to type II plugs. (See 6.1.1.)

3.2.1.2 Type II. Type II plug shall meet all the requirements of the specification and, in addition, shall be so constructed as to be easily taken apart by manual means or by standard wrenches when necessary to refill with fresh desiccant. Each type II plug shall be capable of being disassembledd, refilled and reassembled in 30 seconds. (See 4.6.12, 6.1.2 and table III.)

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3.2.2 Gasket, exterior seating. The gasket shall fit snugly enough so that it will not pass the first thread after subjection to the desiccant retainer test of 4.6.2. (See 4.4.1 and table III.)

3.2.3 Seal plug. The seal shall not interfere with visual inspection of the color of the contained desiccant, and the design shall be such as to prevent installation of the plug until the seal has been removed. Removal of the seal shall be accomplished manually or with the use of common tools. The plug shall remain intact during removal of the seal. (See 4.4.1.)

3.2.4 Retainer (filter). The retainer (filter) shall be designed to conform to the dimensions of the moisture entrance end of the plug, as shown on the applicable MS drawings, and to retain the desiccant. (See 4.4.1 and table III.)

3.3 Plug gasket installation and removal.

3.3.1 Installation. Plugs having screw threads shall be such as to permit the plugs to be screwed into the threaded hole at the seating torque specified on the applicable MS drawing and in table I. The exterior seating gasket shall contact the sealing surface of the engine boss at a maximum torque of 2 pound-inches. Application of the proof torque, specified in table I, shall not cause failure of the material. (See 4.6.1.)

3.3.2 Removal. The thread shall be broken loose from the seat and shall be capable of being removed at a torque not greater than 2 pound-inches. The foregoing installation and removal of the plug shall cause no detrimental effects on the threads of the plug or boss. (See 4.6.1.)

TABLE I. Screw thread size and torque.

MS part no.		Thread size (ref.)	Torque values (lb-in)	
Type I reusable	Type II reusable and refillable		Seating torque (maximum)	Proof torque (minimum)
MS3396-1A MS3396-2	MS3396-11 MS3396-12	14-mm 14-mm	18 18	25 25
MS27215-1	MS27215-2	18-mm	25	50

3.4 Desiccant. Plugs shall be filled with desiccant conforming to MIL-D-3716, type IV, grade H in the amounts specified in table II. (See 4.4.1 and 4.6.3.)

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TABLE II. Quantity of desiccant in plugs.

Dehydrator plug	Type	Grams, minimum
14 mm	I	6
	II	5
18 mm	I	6
	II	5

3.5 Identification of product. Each plug shall be marked with the applicable MS part number and name and address of the manufacturer in accordance with MIL-STD-130. (See 4.4.1.)

3.6 Performance. The dehydrator plugs shall conform to the requirements of table III when tested as specified in Section 4.

3.7 Workmanship. The completed dehydrator plug shall show no mold marks, cracks, splits, other surface and visible defects that may impair the physical properties and performance characteristics of the plug. All details of workmanship shall conform to the best commercial practice covering this type of product. (See 4.4.1.)

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TABLE III. Performance requirements.

Characteristic	Requirements	Test paragraph
Screw thread torque	Per table I.	4.6.1
Gasket, exterior seating: As received	Shall not pass first thread.	4.6.2
Desiccant retainer: As received After aging	Shall not become displaced; only negligible "dusting" of desiccant shall occur.	4.6.2
Visibility of desiccant	Quantity of desiccant in opaque portion of plug shall not exceed 80% of minimum charge specified in table II.	4.6.3
Seal effectiveness	Moisture absorbed: 1.5 mg/g, maximum based on minimum charge specified in table II.	4.6.4
Moisture content of desiccant: as received	6% by weight, maximum.	4.6.5
Watervapor absorption of desiccant	Shall be capable of absorbing 0.28 g of watervapor per 48 hrs per sq. cm. of plug cross section.	4.6.6
Watervapor transmission of plug	Weight of desiccant in plug shall not change more than 1.5 mg per gram, based on minimum charge specified in table II.	4.6.7
Thermal stability and reactivation (plugs)	(a) Shall show no deformation or loss of physical properties. (b) Shall conform to watervapor transmission requirement (4.6.7). (c) Moisture content of desiccant after watervapor transmission test shall not exceed 6% by weight (4.6.5).	4.6.8

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TABLE III. Performance requirements. (Continued)

Characteristic	Requirements	Test Paragraph
Pressure (leakage) 18 mm plugs 14 mm plugs	Unless otherwise specified, plugs shall not leak with the following applicable positive air pressure in psig applied at filter retainer end: 10 psig 2 psig	4.6.9
Corrosion (plugs)	Components, exclusive of plug seal, shall not corrode. Discoloration shall not be considered as corrosion.	4.6.10
Flammability of desiccant retainer	Retainer shall not burn or glow after flame is removed.	4.6.11
Disassembly, refill & assembly type II	Complete the procedure in 30 secs.	4.6.12

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of components and materials. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2 Classification of inspection. This specification provides for quality conformance inspection, which shall consist of the examinations and tests specified in 4.4 and 4.6.

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4.3 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions in MIL-STD-105. For purposes of sampling, an inspection lot shall consist of all the plugs made by the same process, from the same manufacturer and to be delivered at one time (or as noted in 4.5).

4.4 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) in 4.4.3. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of plugs for examination under 4.4.1 and in units of shipping containers for examination under 4.4.2.

4.4.1 Examination of the end item for defects in appearance, construction, identification and workmanship. The sample unit for this examination shall be one plug:

No.	Examine	Defects	Classification
101	Glass	Not clear and transparent. Cracked or broken.	Major
102	Metal	Dented, not deburred.	Major
103	Plastic	Split, cracked or broken.	Major
104	Seal	Missing, over or undersized, difficult to remove. Sealing compound applied.	Major
105	Retainer desiccant	Missing.	Major
106	Threads	Not continuous, broken.	Major
107	Plug	Not correct type and size, as specified. Not filled with desiccant. Not in conformance with design, as applicable.	Major
108	Desiccant	Not blue in color (as received).	Major
109	Gasket, exterior seating	Missing. Not class 4, MIL-R-6855. Not proper hardness. Readily slides over threads.	Major
110	Identification	Missing. Not as specified in 3.5 Not durable. Not legible.	Major

4.4.2 Examination of packaging. Examination shall be made to determine compliance with preservation, packing, and marking requirements of Section 5.

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The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed.

No.	Examine	Defects	Classification
111	Material	Color comparison card missing. PPP-C-96 cans do not have multiple-friction top closure. Size of container not as specified. Packing container not as specified.	Major
112	Workmanship	Quantity of plugs in each container not as specified. Level of preservation and packing not as specified. Incomplete or insecure sealing of container flaps. Bulging or distortion of shipping container. Broken, bruised, crushed or otherwise damaged shipping container, which may result in injury to or loss of contents. Marking: Precautionary marking on individual containers missing. Interior and exterior containers not marked in accordance with MIL-STD-129.	Major
113	Weight	Gross weight in excess of weight limitations of container specification.	Major

4.4.3 Inspection levels and acceptable quality levels for examination. The inspection levels for the purpose of determining the sample size and the acceptable quality levels (AQL's) shall be expressed in percent defective, as follows:

Examination paragraphs	Inspection level	AQL
4.4.1	II	1.5
4.4.2	S-4	6.5

4.5 Inspection level and acceptable quality level for tests. The inspection level for the purpose of determining the sample size and the acceptable quality level (AQL) shall be expressed in percent defective, as follows:

Test paragraph	Inspection level	AQL
4.6	S-3	4.0

The lot size shall not exceed 1200 plugs.

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4.6 Tests.

4.6.1 Screw thread torque.

4.6.1.1 Apparatus. For all plugs having straight threads, the apparatus shall include devices or mechanisms whereby torque can be applied with a minimum of eccentric loading and an aluminum-alloy female boss with a blank end. The female thread of the boss shall correspond to the thread size indicated on the applicable MS drawing and shall be countersunk 45 degress to a depth of 1/16 inch with the initial thread starting 1-1/2 thread turns below the inside edge of the countersunk section. The thread of the 18-1.5 mm, the 14-1.25 mm and all machine screw bosses shall extend at least 1/8 inch further than the length of the threads on the plug. The face of the boss, against which the plug seats, shall be normal to the axis of the threads within ±1/2 degree.

4.6.1.2 Procedure. The plug shall be screwed into the boss by means of the torque-measuring device to determine the installation running torque. The plug shall then be seated to the proof torque. The plug shall remain seated for 4 hours after which it shall be broken from the seat and the removal running torque determined. The threads on the tested indicator plug and female boss shall be examined. (See 3.3 and table I.)

4.6.2 Desiccant retainer and gasket, exterior.

4.6.2.1 Procedure. A plug, with the seal removed, shall be supported in a vertical position with the retainer down, directly over the piece of metal painted black, which is on a hard solid surface. The plug shall then be allowed to drop freely from a height of 6 inches onto the metal surface. This shall be repeated twenty times, and then the metal surface shall be examined for particles of desiccant, and the plug shall be examined to determine if the retainer has become displaced. Examine the gasket, exterior, to determine that it has not moved past the first thread of the plug (see table III and 3.2.2). The test shall be repeated on an unsealed plug, which has been exposed for 7 days to a temperature of 120° F and a relative humidity of 95 percent. (See table III.)

4.6.3 Visibility of desiccant.

4.6.3.1 Specimen. A specimen of desiccant (conforming to MIL-D-3716, type IV, grade H with a maximum water content of 6 percent) equal to 80 percent of the amount specified in the applicable MS drawing shall be prepared.

4.6.3.2 Procedure. A plug shall be taken apart, and the desiccant removed. The prepared specimen quantity shall be transferred to the plug and the plug reassembled. With the plug held in a vertical position with the threaded end downward, the specimen quantity of desiccant shall be examined for conformance to applicable requirement of table III.

4.6.4 Seal effectiveness.

4.6.4.1 Apparatus. The apparatus shall include six stoppered, weighing bottles, a scale capable of weighing to the nearest milligram, a humidity cabinet described in ASTM D 1748 and an oven.

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4.6.4.2 Procedure. The seal shall be taken off six plugs, and all the desiccant removed. An amount of desiccant equal to at least 75 percent of the amount specified as a minimum charge per plug on the applicable MS drawing (see table II), shall be placed in each of six weighing bottles. The weighing bottles shall be stoppered and weighed to the nearest milligram. Removal of the desiccant from the plugs may be accomplished as follows:

(a) Disconnect the glass part from the body of the plug where means are provided for this operation, or

(b) Remove or puncture the retainer in the vented end of the plug.

The seal shall be retained for reinstallation. After weighing, the desiccant shall be rapidly transferred to the plug from which it was removed, and the plug immediately sealed with the original seal. Three of the sealed plugs shall then be exposed in a humidity cabinet to an atmosphere of 120 ±2° F and a relative humidity of 92.5 ± 2.5 percent for a period of 72 hours. The other three shall be exposed in an oven maintained at a temperature of 120 ± 2°F for 72 hours. The desiccent shall then be removed from the plugs and reweighed in the six stoppered weighing bottles to the nearest milligram. The results shall be calculated as follows and shall meet the applicable requirement of table III:

$$\text{Gain (mg/g)} = \frac{\text{humidity cabinet change in mg - average oven pickup in mg}}{\text{minimum charge in grams specified on applicable MS drawing}}$$

4.6.5 Moisture content of desiccant as received.

4.6.5.1 Apparatus. The apparatus shall include three tared, covered, silica crucibles, a scale capable of weighing to the nearest milligram, an electric furnace capable of reaching temperatures of 1750°F and a desiccator.

4.6.5.2 Procedure. Three plugs shall be opened and their desiccant content immediately placed in each of three tared, covered, silica crucibles and weighed with the covers in place. Place the crucibles with cover lids in a cool electric furnace. Allow the temperature to rise to a bright red heat (about 1750°F) in one hour. Remove cover lids from crucibles, and heat for one-half to three quarters of an hour at 1750° F. Remove crucibles from furnace, and place in a desiccator for one-half hour to cool before weighing. Moisture content percentage shall be computed as follows and shall meet the applicable requirement of table III:

$$\frac{\text{Initial wt of desiccant - final wt of desiccant}}{\text{Initial weight of desiccant}} \times 100 = \text{Percentage moisture - content}$$

4.6.6 Watervapor absorption.

4.6.6.1 Apparatus. The apparatus shall include a humidity cabinet described in ASTM D 1748 and all the apparatus required for the moisture content test (see 4.6.5.1).

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4.6.6.2 Procedure.

4.6.6.2.1 Type I plug. Three type I plugs shall be unsealed and immediately suspended in a vertical position with the absorbing end downward in a humidity cabinet at a dry bulb temperature of 120 ±2°F and 92.5 ±2.5 percent relative humidity for 48 hours. At the end of this period, the plugs shall be broken open, all desiccant removed and the weight (W_1) of the desiccant from each plug determined. The same desiccant from each plug shall be dried and weighed (W_2) to determine the moisture content by the procedure used in the moisture content test (4.6.5.2). The moisture content of the desiccant in three additional plugs that are in the "as received" condition shall be determined and averaged (M_o). The actual amount of watervapor pick-up for the three test plugs shall be calculated as follows:

$$\text{Amount of water picked up (grams)} = W_1 - W_2 - \left(\frac{W_2}{W_o}\right) M_o$$

Where M_2 = weight of moisture content of test desiccant prior to test = $\frac{W_2}{W_o} M_o$

W_1 = weight of test desiccent after exposure in humidity cabinet.

W_2 = weight of test desiccant after moisture content test.

M_o = average weight of moisture content "as received" plugs.

W_o = average weight of desiccant after moisture content test for "as received" plugs.

All units are in grams.

The above value is then divided by the area of the plug based on the nominal thread diameter to obtain the watervapor absorption value. The plugs shall meet the applicable requirement of table III.

4.6.6.2.2 Type II plug. The contained desiccant of three type II plugs shall be removed and weighed to the nearest milligram, the desiccant replaced, the seal removed, and the plugs exposed in a humidity cabinet under the same conditions and period of time as for the type I plugs. After exposure, the contained desiccant of each plug shall again be removed and weighed. The watervapor pick-up shall be determined for each plug and shall meet the applicable requirement of table III.

4.6.7 Watervapor transmission of plug.

4.6.7.1 Apparatus. The apparatus shall include six stoppered weighing bottles, a scale capable of weighing to the nearest milligram, six aluminum-alloy bosses of the type described in 4.6.1.1, a humidity cabinet, described in ASTM D 1748 and an oven.

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4.6.7.2 Procedure. The seal shall be taken off six plugs, the desiccant removed, and an amount of desiccant equal to at least 75 percent of the amount specified as a minimum charge on the applicable MS drawing placed in separate weighing bottles, stoppered and weighed to the nearest milligram. Removal of the desiccant from the plugs may be accomplished as follows:

(a) Disconnect the glass part from the body of the plug where means are provided for this operation, or

(b) Remove or puncture the retainer in the moisture entrance end of the plug.

All of the desiccant, previously removed, shall then be rapidly transferred to the plug from which it was removed, and the plug immediately screwed into a boss. The plug shall be screwed into the boss to the seating torque specified in the applicable MS drawing. Three assemblies shall be exposed in a humidity cabinet to an atomsphere of 120 ±2°F and a relative humidity of 92.5 ±2.5 percent for a period of 72 hours. The other three assemblies shall be exposed in an oven maintained at a temperature of 120 ±2°F for a period of 72 hours. At the end of the exposure periods in the humidity cabinet and oven, the plugs shall be detached from the boss, the desiccant removed and the desiccant from each plug reweighed to the nearest milligram in a stoppered weighing bottle. For the plugs exposed in the oven, the average change in weight from the three plugs shall be calculated. The three results of the watervapor transmission test shall be calculated as follows:

$$\text{Gain (mg/g)} = \frac{\text{humidity cabinet change in mg - average oven pickup in mg}}{\text{minimum charge in grams specified in table II}}$$

The watervapor transmission for each plug shall meet the applicable requirement in Table III.

4.6.8 Thermal stability and reactivation.

4.6.8.1 Apparatus. The apparatus shall include a humidity cabinet, described in ASTM D 1748, an oven with two narrow supports and a desiccator.

4.6.8.2 Procedure. A plug with the seal removed shall be placed in a humidity cabinet at a dry bulb temperature of 120 ±2°F and relative humidity of 92.5 ±2.5 percent for 120 hours. At the end of this time, the plug shall be placed in an oven at a temperature of 265 to 275°F in a horizontal position, supported freely on two narrow supports, each support to be located within 1/4 inch of the end of the plug. After 24 hours of such conditioning, the plug shall be removed from the oven and placed in a desiccator and allowed to cool for one hour. The plug shall be examined carefully to determine whether it has been deformed in any way physically by the temperature of the oven. The plug shall then be tested to determine conformance with the requirements for watervapor transmission (see table III and 4.6.7). Following this test, the plug shall be opened, and the moisture content of the contained desiccant determined. The plug shall meet the applicable requirement in table III.

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4.6.9 Pressure (leakage).

4.6.9.1 Apparatus. A boss capable of gripping the base of the plug by screwing into the threaded section, a source of air pressure (gage) and either a container of water or gages suitable for detecting a loss in pressure.

4.6.9.2 Procedure. The plug shall be fitted into a boss by screwing into the threaded section. For the 18 mm cyclinder plugs, 10 psig shall be applied to the filter end of the plug. For 14 mm plugs, 2 psig shall be applied to the filter end of the plug. The assembly shall be examined for leakage in accordance with Method 5009 of FED.TEST METHOD STD. NO. 101. (See table III.)

4.6.10 Corrosion.

4.6.10.1 Procedure. A plug shall be exposed to an atmosphere of 120 +2°F and 100 percent relative humidity in a humidity cabinet described in ASTM D 1748 for a period of 150 hours. At the completion of the exposure period, the plug shall be examined for corrosion. Neither corrosion of the plug seal (provided the corrosion does not interfer with the removal of the seal from the plug) nor discoloration of the metal components of the plug (provided there is no pitting) shall be causes for corrosion failure. (See table III.)

4.6.11 Flammability of desiccant retainer.

4.6.11.1 Procedure. The desiccant retainer or its components from a plug shall be exposed to the flame of a bunsen burner for 5 seconds. The flame shall then be removed, and the retainer or its components examined to determine compliance with the applicable requirement of table III. Light charring shall not be considered cause for rejection.

4.6.12 Disassembly, refill and assembly. A filled type II plug shall be disassembled, disiccant replaced and reassembled either manually or with standard tools in the time stated in table III.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or industrial, as specified (see 6.2).

5.1.1 Level A. The plugs shall be snugly packed, cushioned with chemically inert dunnage, if necessary, in 1-gallon, multiple-friction closure cans conforming to type V, class 2 of PPP-C-96. In each can shall be placed one comparison card conforming to MS3398. When specified by the acquisition activity, containers conforming to MIL-D-6054 may be used as an alternative container. The quantity of plugs preserved in each container shall be as specified by the acquisition activity (see 4.4.2 and 6.2).

5.1.2 Industrial. Plugs shall be preserved in accordance with ASTM D 3951.

5.2 Packing. The levels of packing shall be Level A, B or Industrial as specified (see 6.2).

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5.2.1 Level A. Unless otherwise specified by the acquisition activity, plugs, preserved in accordance with 5.1, shall be packed in containers conforming to overseas type PPP-B-601 or overseas type PPP-B-621 (gross weight not to exceed 150 pounds).

5.2.2 Level B. Plugs, preserved in accordance with 5.1, shall be packed in containers conforming to domestic types PPP-B-601 or PPP-B-621 (gross weight not to exceed 150 pounds) or overseas type PPP-B-636 class (gross weight not to exceed 70 pounds).

5.2.3 Industrial. Plugs, preserved in accordance with 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Interior and shipping containers shall be marked as specified in MIL-STD-129.

5.3.2 Precautionary marking. All PPP-C-96 containers shall be marked with the following precautionary marking:

CAUTION: REPLACE COVER AFTER EACH
PARTIAL REMOVAL OF CONTENTS.

6. NOTES

6.1 Intended use. The reusable plugs covered by this specification are intended for use in the preservation of reciprocating aircraft engines and related equipment.

6.1.1 Type I. The type I plug is intended for use at field stations where facilities to reactivate the contained desiccant are available.

6.1.2 Type II. The type II plug is intended for use were facilities to reactivate the contained desiccant are not available, or where it would be more convenient to refill the deactivated plug with fresh activated desiccant. The type II plug may be reactivated, if desired.

6.1.3 Moisture content. Plugs are satisfactory for use when the total moisture content of the contained desiccant does not exceed 7.5 percent at the time the plug is placed in use.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, size, and MS part number required. (See 1.2 and table I.)

(c) Quantity to be packaged in container. (See 5.1.1.)

(d) Level of preservation and packing required. (See 5.1 and 5.2.)

(e) State if MIL-D-6054 containers are to be used. (See 5.1.1.)

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6.3 Changes form previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Air Force - 82
Navy - OS
DLA - GS

User activity:
Army - AR

Preparing activity:
Navy - AS
(Project No. 6685-0723)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-6131C	Plugs, Dehydrator, Indicating Humidity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify)

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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